



ACADEMIC RESEARCH ASSISTANT USING VECTOR SEARCH AND RAG



A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

The work embodied in the present project report entitled “**ACADEMIC RESEARCH ASSISTANT USING VECTOR SEARCH AND RAG**” has been carried out by the students, **GODFRET T R, GRISH NARAYANAN S, HARI PRAKASH A**, The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

ScholarAI is an AI-powered academic research assistant designed to simplify how researchers analyze and interact with research papers. It uses technologies like Retrieval-Augmented Generation (RAG), natural language processing, and machine learning to provide features such as document summarization, semantic search, paper comparison, and conversational interaction. The system is built using a microservices architecture with a React frontend, Python backend & ML services, along with MongoDB and vector databases for data handling. By automating time-consuming research tasks and improving search accuracy, ScholarAI helps researchers save time, improve productivity, and gain deeper insights from academic content.

Keywords: ScholarAI, Academic Research Assistant, Retrieval-Augmented Generation (RAG), Natural Language Processing (NLP), Machine Learning (ML) Semantic Search.

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SIGNATURE

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LIST OF ABBREVIATIONS

AI	-	Artificial Intelligence
ML	-	Machine Learning
NLP	-	Natural Language Processing
RAG	-	Retrieval-Augmented Generation
LLM	-	Large Language Model
API	-	Application Programming Interface
REST	-	Representational State Transfer
HTTP	-	Hypertext Transfer Protocol
HTTPS	-	Hypertext Transfer Protocol Secure
JSON	-	JavaScript Object Notation
NoSQL	-	Not Only Structured Query Language
CORS	-	Cross-Origin Resource Sharing
UI/UX	-	User Interface / User Experience
PDF	-	Portable Document Format
UUID	-	Universally Unique Identifier
FAISS	-	Facebook AI Similarity Search
SDK	-	Software Development Kit
CLI	-	Command Line Interface
ORM	-	Object Relational Mapping
CI/CD	-	Continuous Integration / Continuous Deployment